

LMU MSBA Capstone Project

Project Title!

Program Engagement Analytics and Capacity Optimization for Community Outreach

Company Name	Irvine Ranch Conservancy
Sponsoring Organization Name	
Sponsoring Organization Address	4727 Portola Parkway, Irvine, CA 92620
Proposal Liaison <i>(person who will work with us on this proposal)</i>	Chris Eljenholm - Program Coordinator (celjenholm@irconservancy.org)
Legal Liaison <i>(person who will be our key point of contact during NDA execution)</i>	Name: David Raetz Title: Vice President and Chief Administration Officer Email: draetz@irconservancy.org
Project Liaison <i>(person who will be our key point of contact during Project execution)</i>	Name: Chris Eljenholm Title: Program Coordinator Email: celjenholm@irconservancy.org
Intern Positions <i>(are summer intern positions available while the student team works on the project)</i>	

1. Project Description

(Provide background and contextual information. Define the analysis problem. Clearly identify project goals and deliverables. State the value this analysis will bring to your Organization. Note that the analysis problem must be stated in terms of a product or result that can be accomplished using some form of descriptive, predictive, or prescriptive analytics by a team of 3-4 students in 15 weeks.) Click or tap here to enter text.

Irvine Ranch Conservancy (IRC) manages approximately 30,000 acres of open space land in collaboration with OC Parks, the City of Irvine, and the City of Newport Beach. Many public facing programs are led by various IRC volunteers and Staff to allow public access into our managed areas. However, these programs can vary greatly in timing and content. The goal of this project is to develop a structured analytic framework that integrates program metadata, attendance, capacity and staffing into predictive and decision-support models that inform outreach planning and resource allocation. Goals and deliverables would consist of a clean, standardized analytic dataset from Let's Go Outside activity exports; exploratory analysis of engagement trends; quantified geographic and activity-based engagement gaps; an interactive dashboard to view trends and identify areas where additional programs might be warranted.

2. Business Impact

(Describe how you believe this project will impact your organization in terms of qualitative improvements, cost savings, ROI etc.)

This project will enable IRC to make effective program decisions based on historical program trends which will affect how Let's Go Outside programs are structured, scheduled, and resourced. By analyzing attendance patterns, capacity utilization, no-show dynamics, walk-up participation, and staffing effort across locations and activity types, the project will provide actionable insights into which programs perform best, where outreach gaps exist, and how program offerings align with community demand. The resulting predictive models and dashboard will support more effective capacity planning, scheduling adjustments, and program mix decisions, helping our team optimize outreach impact.

3. Stakeholders and End Users

(Please identify the end-users who will be the direct beneficiary, using or interacting with the artifacts developed in this project)

The primary users of this system will be IRC staff, specifically from our Community Engagement & Education department who are responsible for designing and leading outreach opportunities. Additionally, results from this analysis will be distributed to our volunteer group, highlighting additional program needs that they might be able to assist with. Other beneficiaries include OC Parks, the City of Irvine, and the City of Newport Beach, who are the landowners of the lands managed by IRC.

4. Data Sources and Description

(Provide information on the data you will share, including variables, type, granularity, time-period, number of records, and approximate total size in bytes. Provide information on how you will share this data, including mode of access, platforms, frequency, and format. Confirm that you will be able to provide access to this data by Mar 17th.)

LetsGoOutside.org is the main website where activities are hosted by IRC and other partner organizations. This site allows users to sign up for activities that they wish to participate in. All of this activity data is stored within Let's Go Outside's databases and can be exported into Excel or CSV format for further analysis. Exports can include things like date, time, activity name, activity type, sub type, location, host organization, private event (yes/no), unlisted event (yes/no), number of volunteers, number of registered participants, number of no-shows, number of walkups, number of children, total guests (including volunteers), number of staff, staff hours, total spaces, etc. This data will likely be shared via SharePoint in an excel format. We would be able to provide access to this data by March 17th.

5. Data Confidentiality / Data Privacy / Data Access Requirements

(Provide information on any specific organizational requirements with respect to Data Confidentiality and Data Privacy, such as, (a) can the data be stored in cloud such as AWS for the project? (b) will the data be needed to be anonymized (c) will the students have access to your organizational data sources? etc.)

Any use of IRC's data outside of this project is strictly prohibited. Any data containing identifiable information must not be shared publicly unless explicitly authorized by IRC staff. Data will remain on internal storage systems unless alternative arrangements, such as cloud storage, are explicitly approved by IRC.

6. Technology Description

(Provide information on any specific Platforms and Tools you that you might require the students to use for data analysis, and if so, confirm that you will be able to provide access to these by Dec 1st. If there aren't any, say "none".)

We primarily work with the Microsoft 365 environment including SharePoint, Excel, and related Microsoft Tools. While we are open to students recommending appropriate platforms for data analysis, we prefer solutions that are compatible with Microsoft products to ensure long-term sustainability and internal usability.

For data analysis and modeling, tools such as Python, R, SQL, or other standard analytics environments are acceptable. For visualization and dashboard development, we have limited prior experience with Power BI and would prefer a solution that can be maintained within Power BI Desktop, which is freely available and should not require additional licensing for local use. If students recommend an alternative dashboard platform, it should be low-cost, sustainable, and reasonably maintainable by IRC staff after project completion.

7. Preferred Analytics Methods & Project Management Methods

(Provide information on any specific methods or techniques you might require the students to use for data analysis and project management (e.g., waterfall, agile etc.). If there aren't any, say "none".)

This project may involve descriptive analytics and predictive modeling. We do not require specific algorithms but expect students to apply appropriate, well-documented analytical techniques to evaluate model performance against reasonable baselines. Therefore, no specific project management methodology is required, and we are comfortable with students exploring potential solutions. Our focus is on determining where programs are offered, what types of activities are most effective, where gaps may exist, and how attendance, no-shows, and walk-ups vary by program type, location and timing.

8. Meeting Schedule

(The MSBA Capstone Projects have the following milestones to ensure the Project stays on course. Identify the person from your organization who will be the primary liaison with the students and faculty. We recommend the liaison meet with the students and faculty advisor on a weekly basis. The liaison will be responsible for providing data that the students will be using for the capstone. The liaison will also attend the kick-off and final presentations).

Meeting	Date (Fridays)	Attendees
Project Proposals Intake	February 24, 2025	MSBA team, Advising Faculty Host organization representatives
Project Kickoff (major data is in)	March 14, 2025	MSBA Team, Advising Faculty Host organization representatives
Weekly Sync	May 19-August 8, 2025	MSBA Team, Advising Faculty (Preferably at least 30 minutes weekly committed by Liaison)
Kick-off Presentation	June 13, 2025	MSBA Team, Advising Faculty Host organization representatives
Final Presentation	August 8, 2025	MSBA Team, Advising Faculty Host organization representatives

9. Deliverables

(Clearly identify key deliverables required of the student team. The MSBA Capstone Projects require a mid-term presentation, a final presentation, and a final report. If you would like additional deliverables, add those to the table.)

(a) LMU Student Team Deliverables:

Deliverable	Date (Fridays)	Description
Kick-off Presentation	June 13, 2025	Problem statement, scope, assumptions, analytic plans, data requirements and next steps
Final Presentation	August 8, 2025	Problem statement, business value, scope, assumptions, methodology, results and conclusions, limitations, possible extensions.
Final Report	August 10, 2025	Final reports will be shared with host organizations.
Software and related documentations	August 10, 2025	Student team will hand over all software code, application, models, output, and related documentations necessary for CHLA to continue or implement the solution arrived at in this project.
<i>(specify additional)</i>		

(b) Client Deliverables:

Deliverable	Date (Fridays)	Description
Student and Faculty Instructor Access to required IT Systems	March 14, 2025	LMU Students and the faculty instructor should be able to access Client IT systems necessary to carry out the project.
Data	April 11, 2025	All required data will be made available to LMU students and faculty instructor. Client will arrange to provide exports of data from the back-end servers when such data is not accessible directly by the project team.
Servers	April 11, 2025	Client will provide necessary server or systems where the data is made available and necessary programming and analytical tools (such as Python, Tableau, Excel etc.) are available or can be installed. Such a server should be accessible to the project team remotely in a development-friendly manner (such as via cloud virtual servers like AWS or Azure or Google Cloud).
Subject Matter Expert	April 11, 2025	Client will engage a subject matter expert with the knowledge of the business operations to the project team to participate in weekly meetings, questions from the students, solution brainstorming, student presentations and feedback.
IT Liaison	March 14, 2025	Client will make a management representative available to the project team to help in coordinating matters involving data, servers, team access, programming tools, data security issues, etc., with the Client corporate IT management.

10. Key Success Criteria

(Clearly list key outcomes expected from the student team in terms of KPI, and business implications. Please also indicate whether the project needs future improvement work or

extensions (i.e., next year's

student team could continue on the project.)

The success of this project will be determined by the development of a structured, refreshable analytic framework that integrates Let's Go Outside program metadata, attendance, capacity, and staffing data into a standardized dataset suitable for ongoing analysis.

Students should develop predictive models related to attendance, capacity utilization, or no-show dynamics. Model performance should be evaluated using appropriate validation techniques.

The project should produce quantified, decision-relevant findings, including identification of at least several geographic, activity-based, or scheduling-related engagement gaps supported by measurable differences in attendance, utilization, or participation patterns.

Final deliverables should include an interactive dashboard or equivalent analytic tool that our community engagement staff can independently use and refresh with updated data exports, along with clear documentation of methodology, assumptions, limitations, and potential areas for future enhancement.

This project may serve as a foundation for future extensions, including integration of participant survey data, demographic analysis, or additional modeling in subsequent years.

11. Additional Comments

(Please share with us any additional comments regarding the project so we can take into consideration when advising the student team.)

Additional Project Contacts:

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